

Table 2. Comparison of HALO with prior antibacterial-specific synergy prediction frameworks

Method	Model / architecture	Feature type	Output	Dataset scale	Evaluation scheme	Reported metrics
INDIGO [17]	Random forest	Chemogenomic fitness profiles	Continuous Loewe α	105–171 pairs, 15–19 cmpds, 1 strain	Leave-drug-out / leave-pair-out	$R = 0.52$ ($P = 10^{-6}$), 66-pair test
CoSynE [16]	Multiple classifiers compared; best type unspecified	Structural substructure profiles	3-class (syn/add/antag)	153 pairs, 18 cmpds, 1 strain	K-fold / leave-drug-out / leave-pair-out	AUROC 0.77, Acc 0.81 (leave-pair-out)
Lv et al. [18]	Graph neural network (architecture unspecified)	Molecular graph structure	Binary	91 pairs, 14 cmpds, 1 strain	Not confirmed	Precision ≈ 0.875 , Acc = 0.90
HALO	LightGBM (gradient-boosted trees)	Multi-level CC bioactivity embeddings	Binary	2,449 interactions, 97 cmpds, 10 strains	Nested pair-held-out CV	ROC-AUC 0.78, Accuracy 0.70 (Sec. 2.2)